A **Mini** Project report

On

Lung Cancer Risk Prediction Using Machine Learning

Submitted to Manipal University, Jaipur

towards the partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

in

Computer Science and Engineering

Jan – May 2027

By:

Nishtha Pandey

23FE10CSE00774

MANIPAL UNIVERSITY JAIPUR

INSPIRED BY LIFE

Under the guidance of:

Dr Arpita Banoria

Department of Computer Science and Engineering

Manipal University Jaipur

Index:

Chapter 1: Introduction

1.1 Problem Statement

Lung cancer is one of the most dangerous diseases and is responsible for a large number of deaths across the world. One of the major challenges in dealing with lung cancer is that its symptoms often appear clearly only in later stages, when treatment becomes more difficult and less effective. Because of this, early risk identification is extremely important. If a person's risk level can be estimated earlier using available health and lifestyle information, then doctors and healthcare systems can take preventive or diagnostic action at the right time.

This project focuses on building a machine learning model to predict whether a person is at risk of lung cancer based on multiple factors such as age, smoking habits, passive smoking exposure, respiratory symptoms, medical history, and other health-related indicators. The problem is treated as a binary classification task, where the model predicts whether the lung cancer risk is present or not present. The aim is not to replace a doctor, but to create a data-driven system that can support early screening and risk assessment.

In this project, a dataset containing 5,000 records and 30 attributes was used. These attributes include both lifestyle-related factors, such as cigarettes per day and alcohol consumption, and clinical indicators, such as oxygen saturation, FEV1, CRP level, and x-ray abnormality. By analyzing these variables and training different classification algorithms, the project attempts to find the model that can most accurately classify lung cancer risk.

1.2Importance of the Topic

The topic of lung cancer risk prediction is highly important because lung cancer remains a serious global health concern. Many people are diagnosed only after the disease has progressed significantly, which reduces the chances of successful treatment. If a reliable predictive model can identify high-risk individuals earlier, it may help in encouraging timely medical tests, monitoring, and preventive care.

This topic is also important from a data science perspective. Healthcare datasets often contain many interconnected variables, and machine learning can help uncover patterns that may not be obvious through manual analysis. A prediction model based on patient history and clinical measurements can be useful for screening support, hospital analytics, and health awareness systems.

Another reason this topic is significant is that it combines medical relevance with practical machine learning techniques. In this project, the problem was explored not only through one model, but by comparing several classification methods such as Logistic Regression, KNN, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, and SVM. This makes the study stronger because it shows how different algorithms behave on the same dataset.

Finally, this topic is useful academically because it demonstrates the complete workflow of a machine learning project: data loading, preprocessing, exploratory analysis, model building, evaluation, comparison, tuning, and interpretation of results. Therefore, the project is not only important because of the health issue it addresses, but also because it provides a meaningful and practical application of machine learning in the healthcare domain.

If you want, I can now also expand Methodology / Working into a more report-style 1 to 2 page section with subheadings like Data Collection, Preprocessing, Model Building, and Evaluation Metrics.

Chapter 2: Proposed Methodology

The project was completed in the following steps:

Step 1 : Data Collection and Loading

The dataset was downloaded using kagglehub from the Kaggle lung cancer prediction dataset. It was then loaded into a Pandas DataFrame for analysis.

Step 2 : Data Understanding

The structure of the dataset was examined using:

df.head()
df.info()
df.describe()
df.columns

This showed that:

The dataset has 5,000 rows and 30 columns
All features are numeric
There are no missing values.
The target variable is lung_cancer_risk

Step 3 : Exploratory Data Analysis

EDA was performed to understand the relationships between variables:

1. Pairplots were used for selected features like age, pack_years, crp_level, and lung_cancer_risk
2. Boxplots were used to inspect outliers in all columns ,unique values were checked for every feature
3. A correlation heatmap was created
4. The correlation analysis showed that features such as pack_years, crp_level, cigarettes_per_day, smoking_years, xray_abnormal, and shortness_of_breath were strongly related to lung cancer risk.

Step 4 : Data Preprocessing

The target variable lung_cancer_risk was separated from the input features. The dataset was split into training and testing sets. In the improved part of the project, the data was further split into:

training set
validation set
test set

This was done so that threshold tuning could be performed on validation data instead of the test data. For feature scaling, RobustScaler was used because it is less sensitive to outliers than standard scaling methods.

Step 5 : Baseline Model

The first model used was LogisticRegression. This was chosen because:

- it is a strong baseline for binary classification
- it is simple and interpretable
- it performs well on structured tabular data

After training the logistic regression model, predictions were evaluated using:

- accuracy
- classification report
- confusion matrix
- ROC-AUC score

Step 6 : Threshold Tuning

Instead of only using the default classification threshold of 0.5, a custom threshold was also tested. In the original part of the notebook, a threshold of 0.3 was applied to improve recall for the positive class. In the improved workflow, threshold optimization was done properly using the validation set.
This is useful because in medical-risk prediction, missing a positive case can be more serious than producing a small number of false positives.

Step 6 : Multi Model Comparision

To make the project stronger, multiple classification models were tested:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- Extra Trees
- Gradient Boosting
- Support Vector Machine

Each model was evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Cross-Validation ROC-AUC

This helped compare both simple and advanced models.

Step 6 : Hyperparameter Tuning

A tuned ExtraTreesClassifier was created using GridSearchCV. Different values of:

- n_estimators
- max_depth
- min_samples_leaf
- max_features

were tested. The best model was selected using ROC-AUC as the scoring metric.

Results/Outputs

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.994	0.9880	0.9880	0.9880	0.9999
SVM	0.983	0.9793	0.9518	0.9654	0.9987
Gradient Boosting	0.972	0.9585	0.9277	0.9429	0.9944
Extra Trees	0.965	0.9421	0.9157	0.9287	0.9944
Random Forest	0.965	0.9421	0.9157	0.9287	0.9936
KNN	0.957	0.9640	0.8594	0.9087	0.9876
Decision Tree	0.949	0.9267	0.8635	0.8940	0.9699

The tuned Extra Trees model gave:

Accuracy: 0.970
Precision: 0.9544
Recall: 0.9237
F1 Score: 0.9388
ROC-AUC: 0.9961

After threshold optimization on the tuned Extra Trees model:

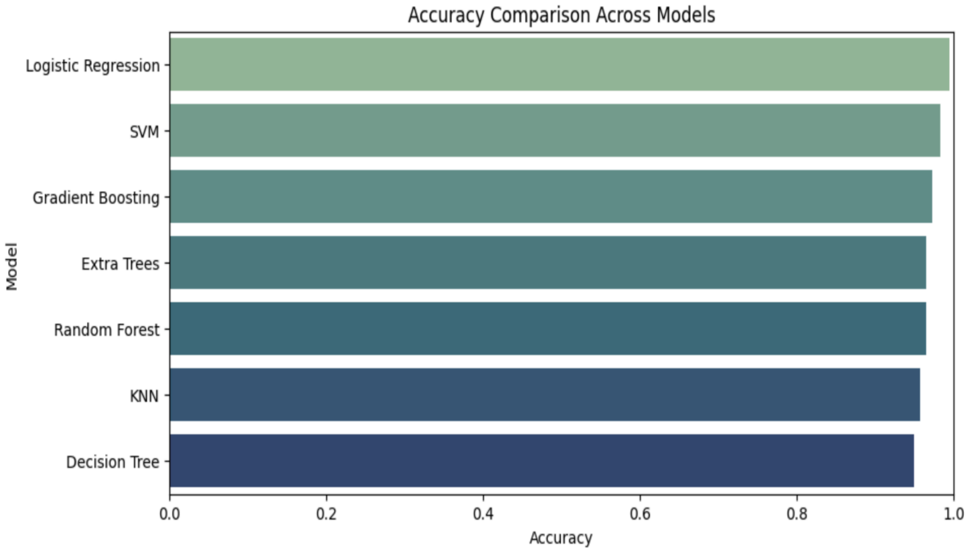
Accuracy improved from 0.971 to 0.972
Recall improved from 0.9277 to 0.9317
F1 Score improved from 0.9409 to 0.9431

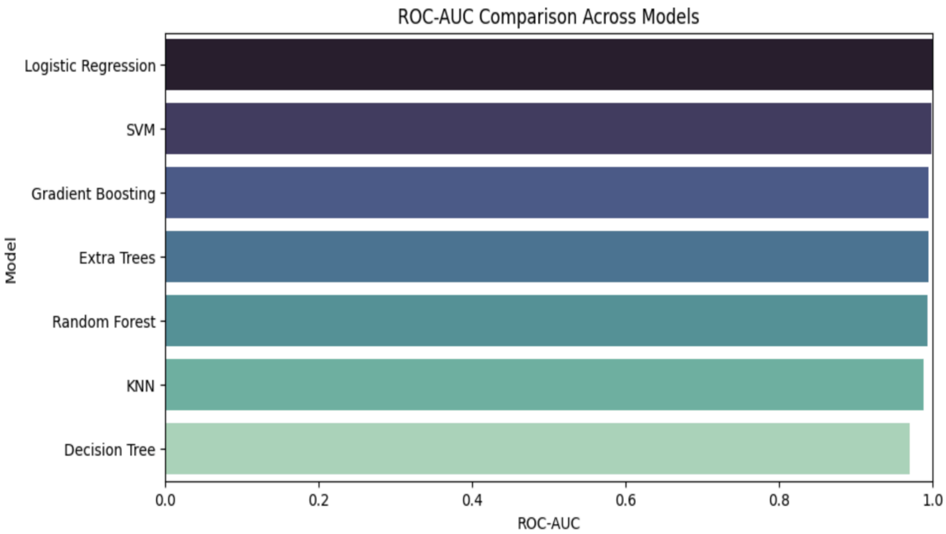
However, the final comparison showed that Logistic Regression remained the best model overall.

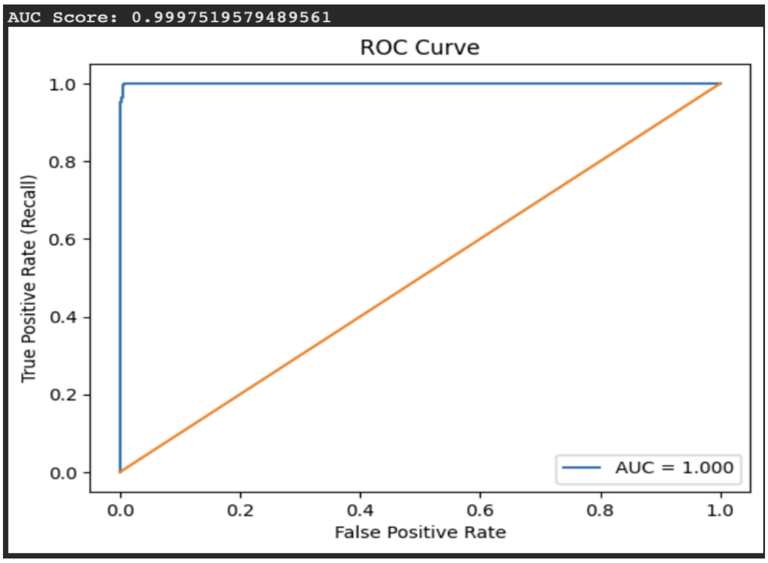
Important interpretation:

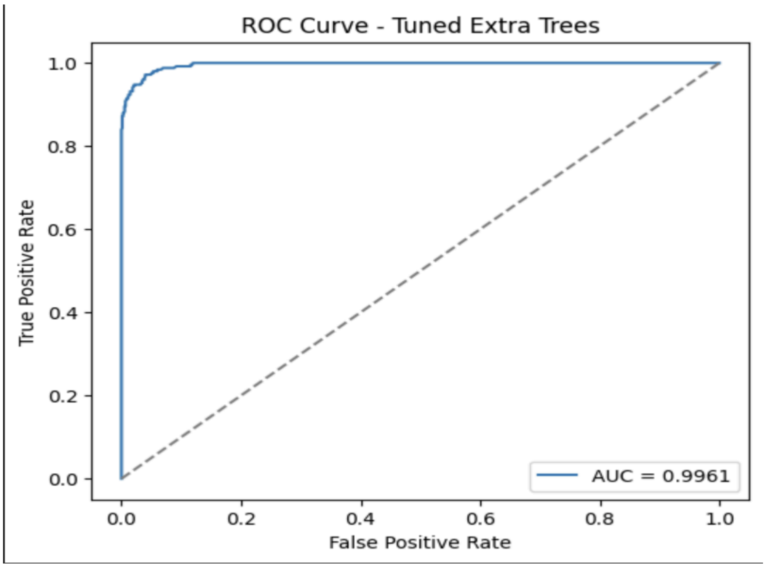
More complex models did not outperform the simple baseline.
This suggests the dataset is highly structured and easily separable.
Therefore, a simpler model was sufficient and more appropriate.

```
sns.barplot(data=results_df, y="Model", x="Accuracy", palette="crest")
```






AUC Score: 0.9997519579489561
ROC Curve
True Positive Rate (Recall)
False Positive Rate
AUC = 1.000
0.0
0.2
0.4
0.6
0.8
1.0


ROC Curve - Tuned Extra Trees
True Positive Rate
False Positive Rate
AUC = 0.9961
0.0
0.2
0.4
0.6
0.8
1.0

Chapter 3: Conclusion

This project successfully predicted lung cancer risk using machine learning techniques. After performing data analysis, preprocessing, and model training, Logistic Regression emerged as the best-performing model with very high accuracy and ROC-AUC. Additional classifiers such as Random Forest, Extra Trees, Gradient Boosting, KNN, SVM, and Decision Tree were also tested, but none performed better than the baseline logistic regression model. Threshold tuning and hyperparameter tuning improved some models slightly, especially recall, but the simple model remained the strongest. This project shows that model comparison is important, and that a simpler model can sometimes outperform more advanced models when the dataset is highly separable.

Future Score

In the future, this project can be improved by testing the model on real-world clinical datasets and by developing a web-based interface for lung cancer risk prediction. More advanced explainability methods such as SHAP can also be added.

Chapter 4: References

[1] KaggleHub Lung Cancer Prediction Dataset.

[2] Shermis, M.D.,&Hamner, B.(2013). Contrasting state of the art automated scoring of essay : PEG and e-Rater

[3] Scikit-learn Documentation.KNeighborsRegressor & KNeighborsClassifier

[4] Manning, C., Raghavan, P., & Schutze, H.(2008). Introduction to information retrieval. Cambridge university press.

[5] Attali, Y.,7 Burstein, J.(2006). Automated essay scoring with e-Rater V.2.

[6] Williamson, D.M., XI,X.,&Breyer,F.J.(2012).A framework for evaluation and use of automated scoring . Educational measurement :issues and practice ,31(1),2-13

- offers evaluation parameters and validation approaches for scoring